

COAL LAB # 02

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COAL LAB

QUESTION NO:01

Write an instruction to:

Move value 5 into AX

Move AX into BX

Add BX and AX

ANSWER:

➤ **Move Value 5 into**

MOV AX, 5

➤ **Move AX into BX**

MOV BX, AX

➤ **Add BX and AX**

ADD AX, BX

QUESTION NO:02

write output:

MOV AX, 5

MOV BX, 3

ADD AX, BX

SUB AX, 2

what will be the final output of AX and BX?

ANSWER:

Step-by-Step Execution:

1. MOV AX, 5

AX = 5

2. MOV Bx, 3

BX = 3

3. ADD AX, BX

AX = AX + BX

$$AX = 5 + 3 = 8$$

4. SUB AX, 2

$$AX = AX - 2$$

$$AX = 8 - 2 = 6$$

Final Output:

$$AX = 6$$

$$BX = 3$$

QUESTION NO:03

What are the differences between RAM and registers?

RAM	REGISTERS
It is the main memory used to store program and data temporarily.	It is used to Store immediate Data for processing.
It is indirectly accessed by CPU.	It is Directly accessed by CPU/ALU.
It has large size (GBs).	It has small size (Bytes).
These are slower than registers.	They are extremely fast.
It is the main memory of computer.	It is located inside the CPU
Example: 10 GB, 20 GB RAM	Example: AX, BX, CX, DX Registers